

PH102

After Mid Semester Exams

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Chapter-9

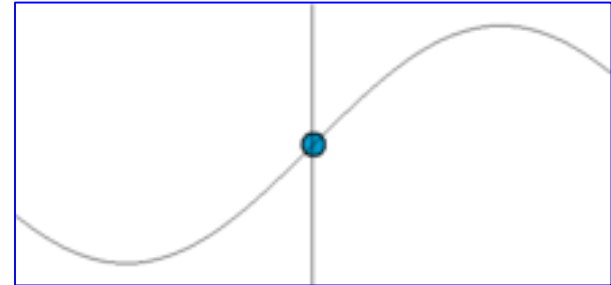
Electromagnetic Waves in Vacuum

General Introduction of Waves

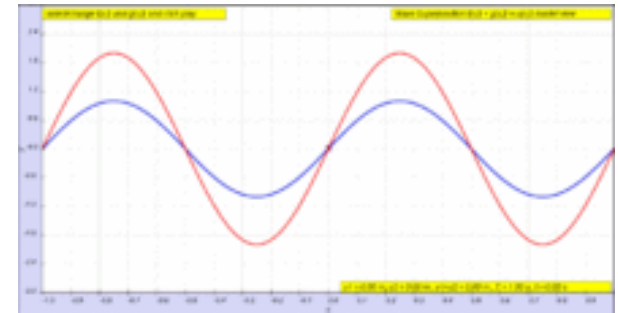




Sinusoidal waves correspond to simple harmonic motion.

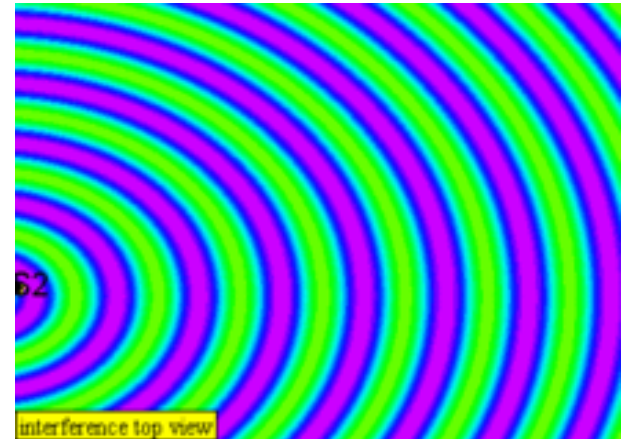
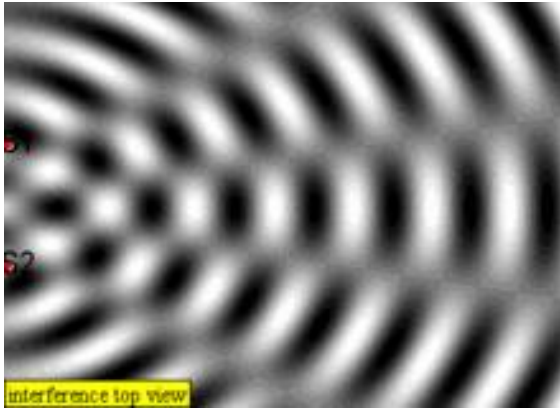


Surface waves in water



Animation for 2 wavelength, green wave traverse to the right while blue wave transverse left, the net red wave amplitude at each point is the sum of the amplitudes of the individual waves. note that $f(x,t) + g(x,t) = u(x,t)$

Ripple tank with two sources of waves interfering each other (Constructive = White and Destructive = Black visualization)

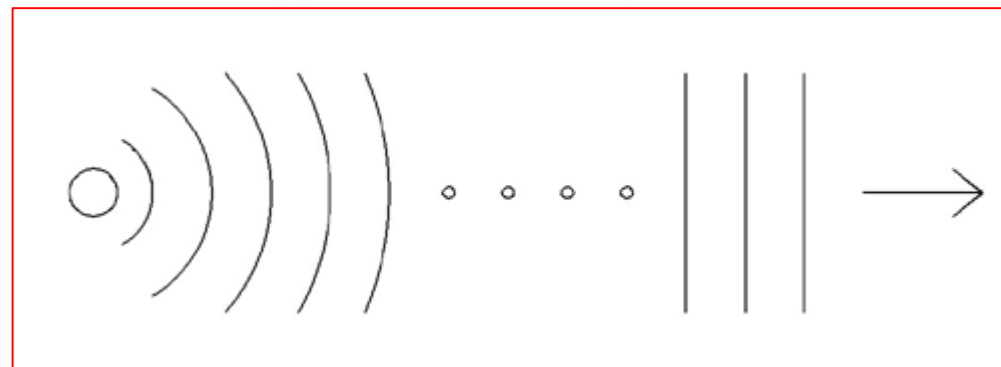
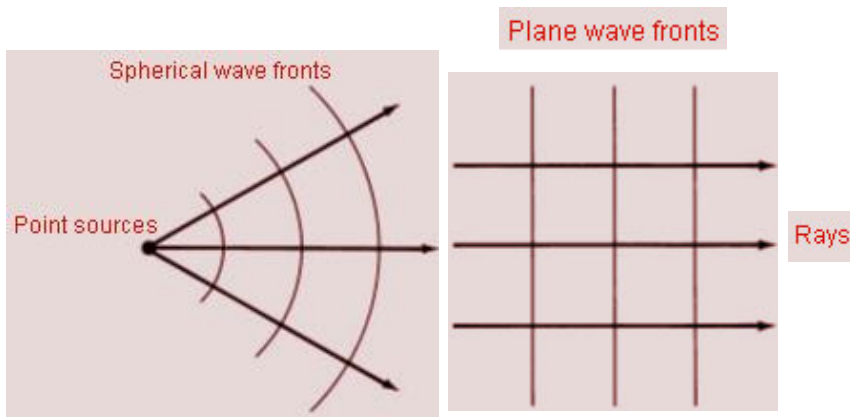


Ripple tank with a spherical source producing circular waves

In 1678, Dutch physicist, Christiaan Huygens proposed that light was made up of waves vibrating up and down perpendicular to the direction of the light travels.

1803, Thomas Young studied the Interference of waves
1815, Augustin Fresnel supported Young's experiments with mathematical calculations.

Plane wave approximation for spherical disturbance

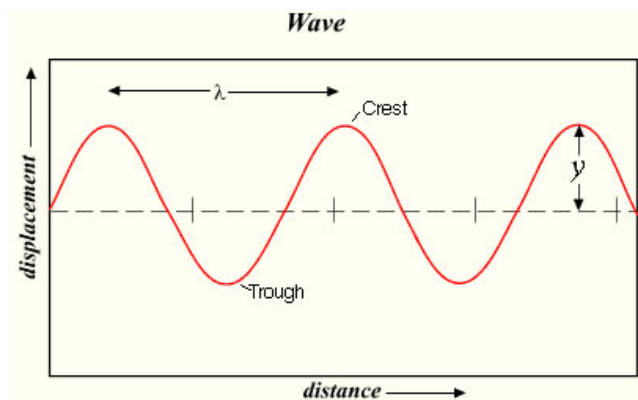


Electromagnetic Waves

Waves in One Dimension

What is a "wave" ?

A wave is a disturbance of a continuous medium that propagates with a fixed shape at constant velocity.



λ = Wavelength
 A = Amplitude

The Wave Equation

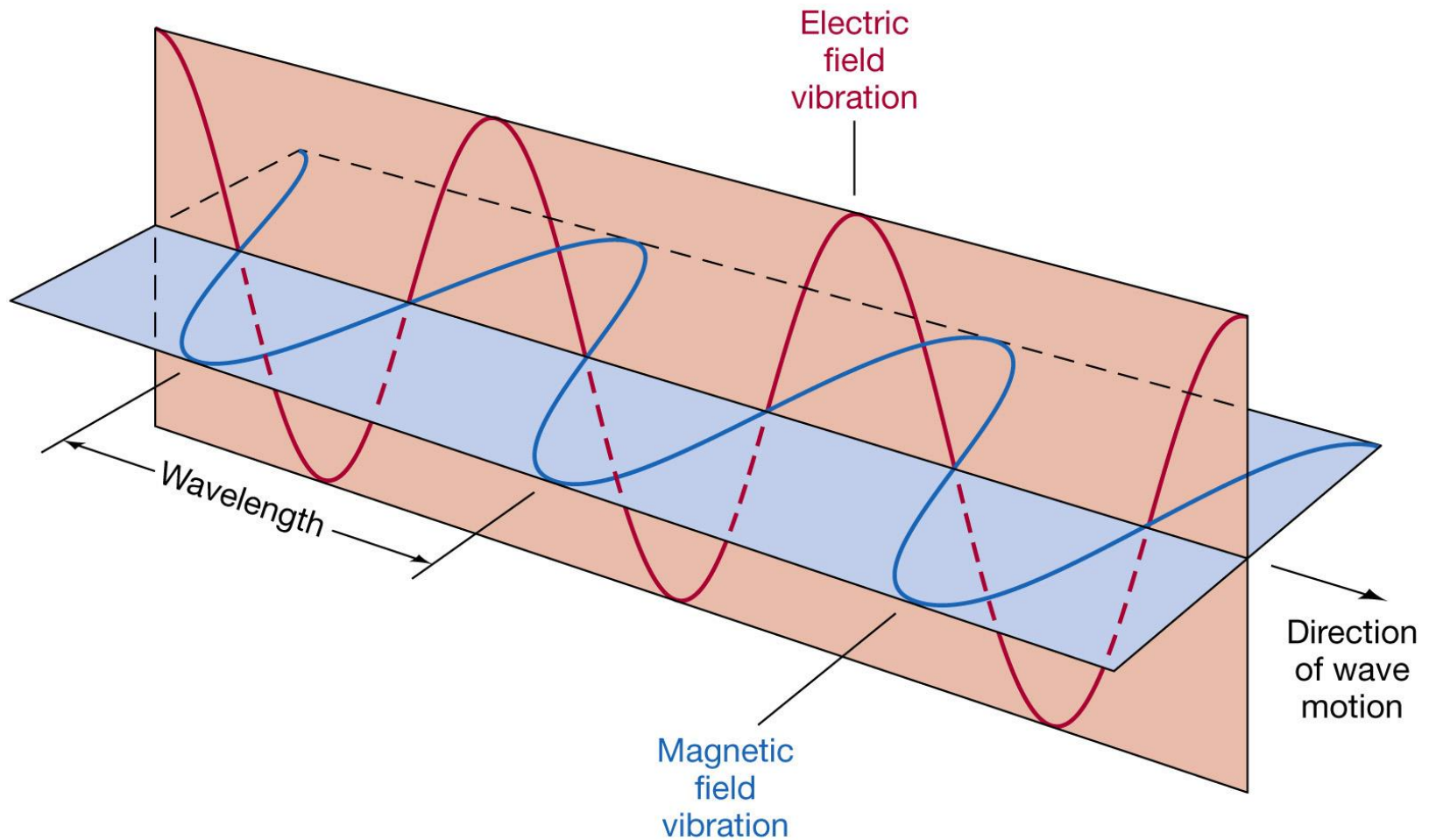
$$\frac{\partial^2 f}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 f}{\partial t^2}$$

f = Displacement of the string

In the presence of absorption, the wave will diminish in size as it moves.

If the medium is dispersive (spread) different frequencies travel at different speeds; in two or three dimensions, as the wave spreads out its amplitude will decrease; and of course standing waves don't propagate at all.

Light = Electromagnetic "waves"



Transverse EM-waves: E & B variations are perpendicular to the direction of propagation

Waves in 1-Dimension

A (classical) 1-D *traveling* wave is a quasi-coherent collective phenomenon – that of a “disturbance” associated with a localized excess of energy (above ambient thermal/background energy) in a macroscopic, continuous medium, which propagates (*i.e.* translates in 1-D space) as time progresses.

In a dissipation-less (*i.e.* **lossless**), non-dispersive medium, the shape/profile/envelope (*i.e.* **crests and troughs**) of the wave propagates with **constant velocity**.

In a dispersive (spread) medium, a traveling wave consisting of a linear combination of several/many different frequencies, the various frequency components of the wave will propagate with different speed, thus the overall shape of the wave will change with time in a dispersive medium.

In a dissipative (but non-dispersive) medium the wave amplitude(s) will decrease with time (or equivalently propagation distance). Often, real media are not only dissipative, but also dispersive, thus dissipation in a medium may also be frequency dependent.

Classical media can be both dispersive and dissipative – one or both or neither.

A (classical) 1-D *standing* wave = a linear superposition of two counter-propagating traveling waves (*e.g.* a standing wave on a stringed instrument.)

Standing waves do not propagate in space, although they can/do evolve in time (due to dispersion, dissipation and other processes).

For non-dissipative, non-dispersive media, the transverse shape {= transverse displacement from equilibrium shape of string} of a traveling wave $f(z, t)$ is invariant under space translations and time translations.

Complex notation.

A sinusoidal oscillation of wave number k and (angular) frequency ω traveling to the left would be written

In view of Euler's formula

$$e^{i\theta} = \cos \theta + i \sin \theta$$

The sinusoidal wave equation, $f(z, t) = A \cos[k(z - vt) + \delta]$, can be written as

$$f(z, t) = \text{Re}[Ae^{i(kz - \omega t + \delta)}]$$

This invites us to introduce the “Complex Wave Function”.

$$\tilde{f}(z, t) \equiv Ae^{i\delta} e^{i(kz - \omega t)}$$

$$\tilde{f}(z, t) \equiv \tilde{A} e^{i(kz - \omega t)}$$

The complex amplitude $\tilde{A} = Ae^{i\delta}$; absorbing the phase constant.

Linear combinations of sinusoidal waves

Although the sinusoidal function is a very special wave form, the fact is that any wave can be expressed as a linear combination of sinusoidal ones:

$$\tilde{f}(z, t) = \int_{-\infty}^{+\infty} \tilde{A}(k) e^{i(kz - \omega t)} dk$$

The combined wave still has the same frequency and wavelength.

Here ω is a function of k ($= \omega/v$), and we have allowed k to run through negative values in order to include waves going in both directions.

The formula for $\tilde{A}(k)$ in terms of the initial conditions $f(z, 0)$ and $\dot{f}(z, 0)$, can be obtained from the theory of Fourier transforms.

Polarization (Wave-Polarization)

The waves that travel down a string when you shake it are called **transverse**, because the **displacement is perpendicular to the direction of propagation**.

If the string is reasonably elastic, it is also possible to stimulate compression waves, by giving the string little tugs (jerks). Compression waves are hard to see on a string, but if you try it with a slinky they're quite noticeable

These waves are called longitudinal, because the displacement from equilibrium is along the direction of propagation. Sound waves, which are nothing but compression waves in air, are longitudinal; electromagnetic waves, as we shall see, are transverse.

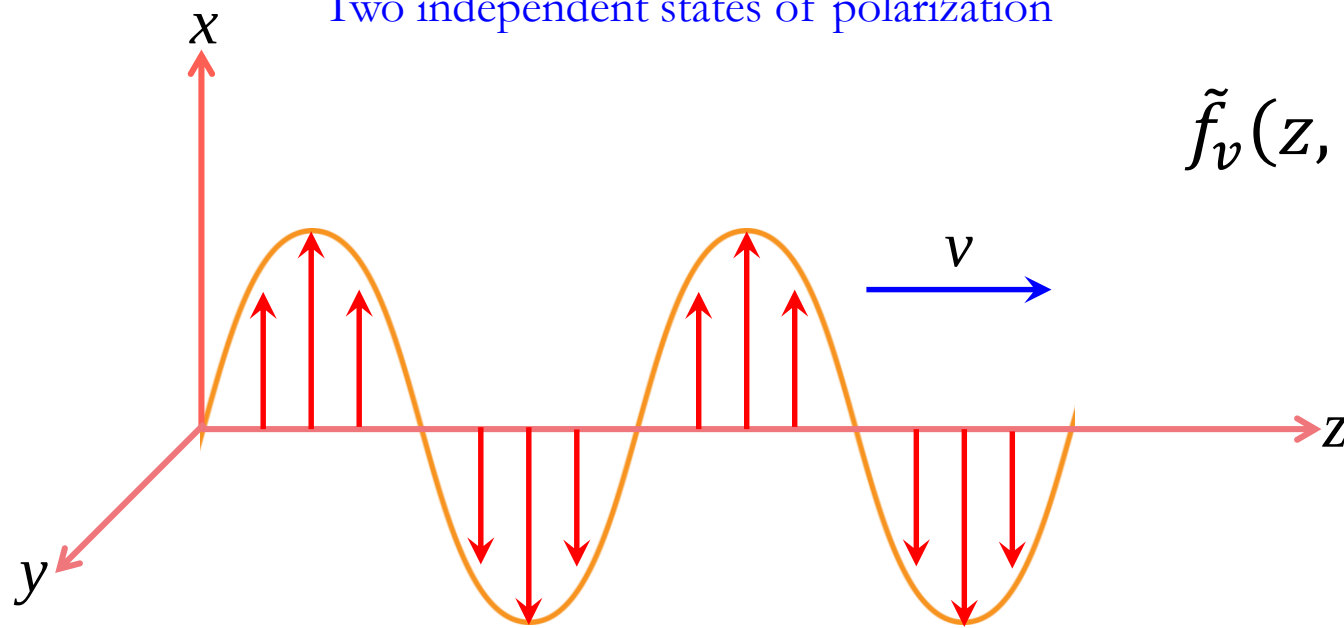
Now there are, of course, two dimensions perpendicular to any given line of propagation. Accordingly, transverse waves occur in two independent states of polarization: you can shake the string up-and-down.

$$\tilde{f}_v(z, t) = \tilde{A} e^{i(kz - \omega t)} \hat{x}$$

Two independent states of polarization

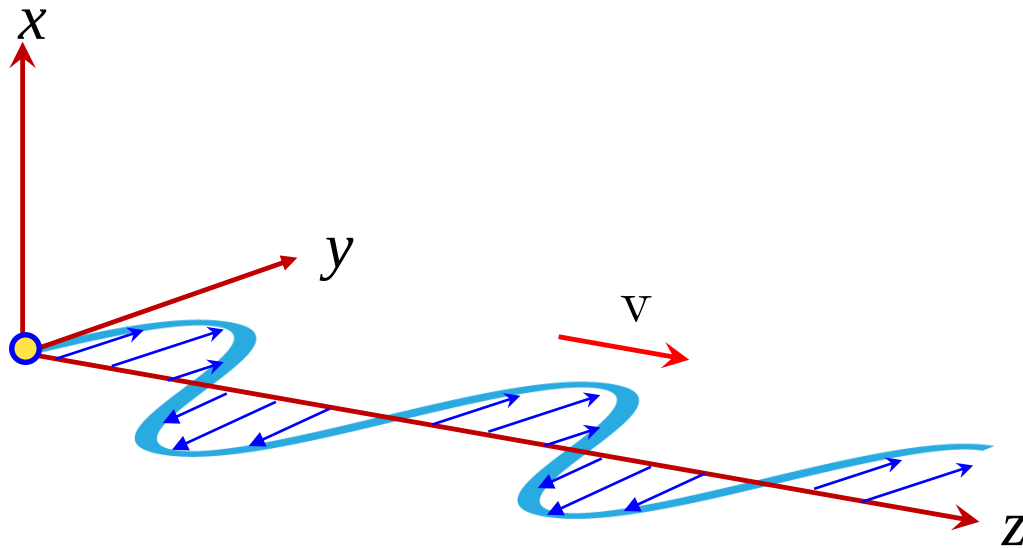
Vertical polarization

$$\tilde{f}_v(z, t) = \tilde{A}e^{i(kz - \omega t)}\hat{x}$$



Horizontal polarization

$$\tilde{f}_h(z, t) = \tilde{A}e^{i(kz - \omega t)}\hat{y}$$



or left-and-right ("horizontal" polarization; Fig. 1.2),

$$\tilde{f}_h(z, t) = \tilde{A}e^{i(kz - \omega t)}\hat{y}$$

or along any other direction in the **xy** plane (Fig. 1.3):

$$\tilde{f}(z, t) = \tilde{A}e^{i(kz - \omega t)}\hat{n}$$

The polarization vector $\hat{n}$ defines the plane of vibration because the waves are transverse, $\hat{n}$ is perpendicular to the direction of propagation:

$$\hat{n} \cdot \hat{z} = 0$$

In terms of the polarization angle θ

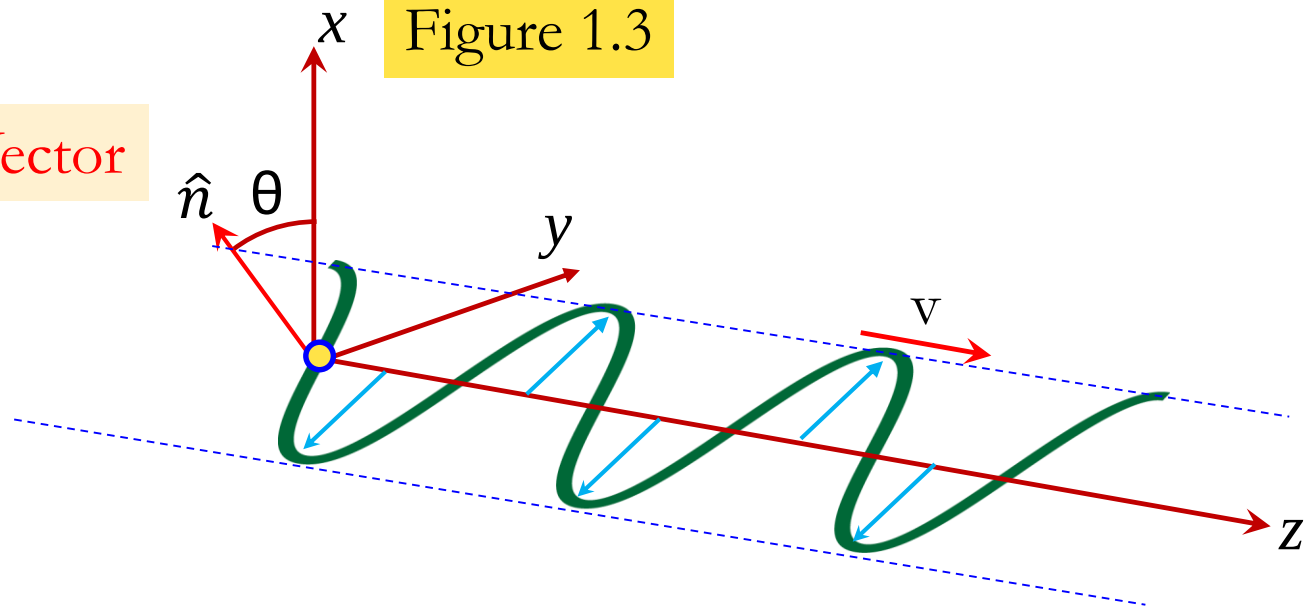
$$\hat{n} = \cos \theta \hat{x} + \sin \theta \hat{y}$$

Thus, the wave pictured in Fig. 1.3 can be considered as a superposition of two waves one horizontally polarized, the other vertically:

$$\tilde{f}(z, t) = (\tilde{A} \cos \theta) e^{i(kz - \omega t)} \hat{x} + (\tilde{A} \sin \theta) e^{i(kz - \omega t)} \hat{y}$$

Figure 1.3

Polarization Vector



Why to use Complex Notation?

Whenever we have two or more waves, if there exists a definite phase relation (defined at some specific origin of time $t = t_0$) between them (*i.e.* the two or more waves are coherent) then the waves will *interfere* with each other at a given point in space, z .

Interference phenomena occurs at the *amplitude* level – *i.e.* wave amplitudes interfere for waves that are coherent/have a (well defined) definite phase-relation.

Using complex notation, phase information can be explicitly {and easily} obtained.

Example: Suppose we want to combine two sinusoidal waves

$$f_3 = f_1 + f_2$$

$$= \text{Re}(\tilde{f}_1) + \text{Re}(\tilde{f}_2)$$

$$= \text{Re}(\tilde{f}_1 + \tilde{f}_2) = \text{Re}(\tilde{f}_3)$$

$$\tilde{f}(z, t) \equiv \tilde{A}e^{i(kz - \omega t)}$$

$$f(z, t) = \text{Re}[\tilde{f}(z, t)]$$

$$f(z, t) = \text{Re}[Ae^{i(kz - \omega t + \delta)}]$$

We simply add the corresponding complex wave functions, and then take the real part.
In particular, if they have the same frequency and wave number,

$$\tilde{f}_3 = \tilde{A}_1 e^{i(kz - \omega t)} + \tilde{A}_2 e^{i(kz - \omega t)}$$

$$= \tilde{A}_3 e^{i(kz - \omega t)}$$

$$\tilde{A}_3 = \tilde{A}_1 + \tilde{A}_2$$

Evidently, you just add the (complex) amplitudes.

$$A_3 e^{i\delta_3} = A_1 e^{i\delta_1} + A_2 e^{i\delta_2}$$

The combined wave still has the same frequency and wavelength.

$$f_3(z, t) = A_3 \cos(kz - \omega t + \delta_3)$$

↓

↓

$$(A_3)^2 = (A_3 e^{i\delta_3}) (A_3 e^{-i\delta_3})$$

$$= (A_1 e^{i\delta_1} + A_2 e^{i\delta_2}) (A_1 e^{-i\delta_1} + A_2 e^{-i\delta_2})$$

$$= (A_1)^2 + (A_2)^2 + A_1 A_2 (e^{i\delta_1} e^{-i\delta_2} + e^{-i\delta_1} e^{i\delta_2})$$

$$= (A_1)^2 + (A_2)^2 + A_1 A_2 2 \cos(\delta_1 - \delta_2)$$

$$A_3 = \sqrt{(A_1)^2 + (A_2)^2 + 2A_1 A_2 \cos(\delta_1 - \delta_2)}.$$

$$A_3 e^{i\delta_3} = A_3 (\cos \delta_3 + i \sin \delta_3)$$

$$A_3 e^{i\delta_3} = A_1 e^{i\delta_1} + A_2 e^{i\delta_2}$$

$$= A_1 (\cos \delta_1 + i \sin \delta_1) + A_2 (\cos \delta_2 + i \sin \delta_2)$$

$$\tan \delta_3 = \frac{A_3 \sin \delta_3}{A_3 \cos \delta_3} = \frac{A_1 \sin \delta_1 + A_2 \sin \delta_2}{A_1 \cos \delta_1 + A_2 \cos \delta_2} \quad \delta_3 = \tan^{-1} \left(\frac{A_1 \sin \delta_1 + A_2 \sin \delta_2}{A_1 \cos \delta_1 + A_2 \cos \delta_2} \right)$$

$$\cos x = \operatorname{Re}\{e^{ix}\} = \frac{e^{ix} + e^{-ix}}{2}$$

$$\sin x = \operatorname{Im}\{e^{ix}\} = \frac{e^{ix} - e^{-ix}}{2i}$$

Electromagnetic Waves in Vacuum

Deriving the Wave Equation for E and B

In regions of space where there is no charge or current, Maxwell's equations read

$$\nabla \cdot \vec{E} = 0$$



1

$$\nabla \cdot \vec{B} = 0$$



2

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$



3

$$\nabla \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$



4

They constitute a set of coupled, first-order, partial differential equations for E and B.

They can be decoupled by applying the curl to eq. (3) and (4):

$$\nabla \times \nabla \times \vec{E} = -\frac{\partial(\nabla \times \vec{B})}{\partial t}$$

$$\nabla \times \nabla \times \vec{E} = -\frac{\partial(\nabla \times \vec{B})}{\partial t}$$

$$\nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\frac{\partial(\nabla \times \vec{B})}{\partial t}$$

$$\nabla \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\frac{\partial(\nabla \times \vec{B})}{\partial t} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\frac{\partial(\nabla \times \vec{B})}{\partial t}$$

no charge or current

since $\nabla \cdot \vec{E} = 0$

$$\nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\nabla \times \nabla \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial(\nabla \times \vec{E})}{\partial t}$$

$$\nabla(\nabla \cdot \vec{B}) - \nabla^2 \vec{B} = \mu_0 \epsilon_0 \frac{\partial(\nabla \times \vec{E})}{\partial t}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

3

$$\mu_0 \epsilon_0 \frac{\partial(\nabla \times \vec{E})}{\partial t} = -\mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2}$$

$$\nabla(\nabla \cdot \vec{B}) - \nabla^2 \vec{B} = -\mu_0 \epsilon_0 \frac{\partial(\nabla \times \vec{E})}{\partial t}$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla^2 \vec{B} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2}$$

$$\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\nabla^2 \vec{B} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2}$$

Equation. 2

We now have separate equations for $\vec{E}$ and $\vec{B}$, but they are of second order; that's the price you pay for decoupling them.

In vacuum, then, each Cartesian component of $\vec{E}$ and $\vec{B}$ satisfies the three-dimensional wave equation,

$$\nabla^2 f = \frac{1}{v^2} \frac{\partial^2 f}{\partial t^2}$$

So Maxwell's equations imply that empty space supports the propagation of electromagnetic waves, traveling at a speed

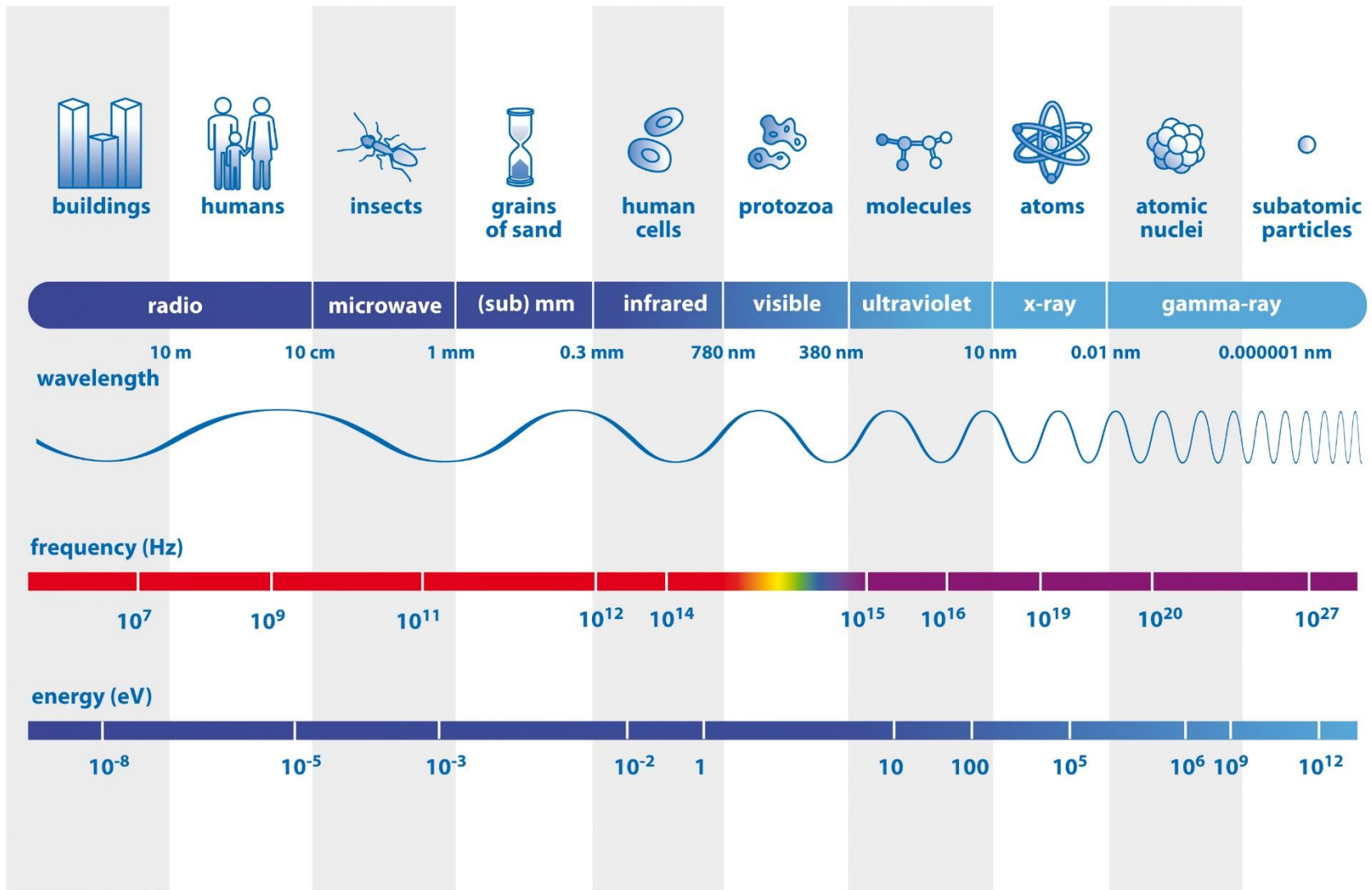
$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/s}$$

This is nothing but precisely the velocity of light, **c**. The implication is astounding: Perhaps light is an electromagnetic wave.

EM waves have associated with them a frequency 'n' and wavelength λ , related to each other via $c = n \lambda$. At the microscopic level, EM waves consist of large numbers of {massless} **real** photons, each carrying energy $E = hn = hc / \lambda$, linear momentum $p = h / \lambda = hn / c = E / c$ and angular momentum $|l_z| = 1\hbar$ where h = Planck's constant = 6.626×10^{-34} Joule-sec and $\hbar = h / 2\pi$

Electromagnetic spectrum



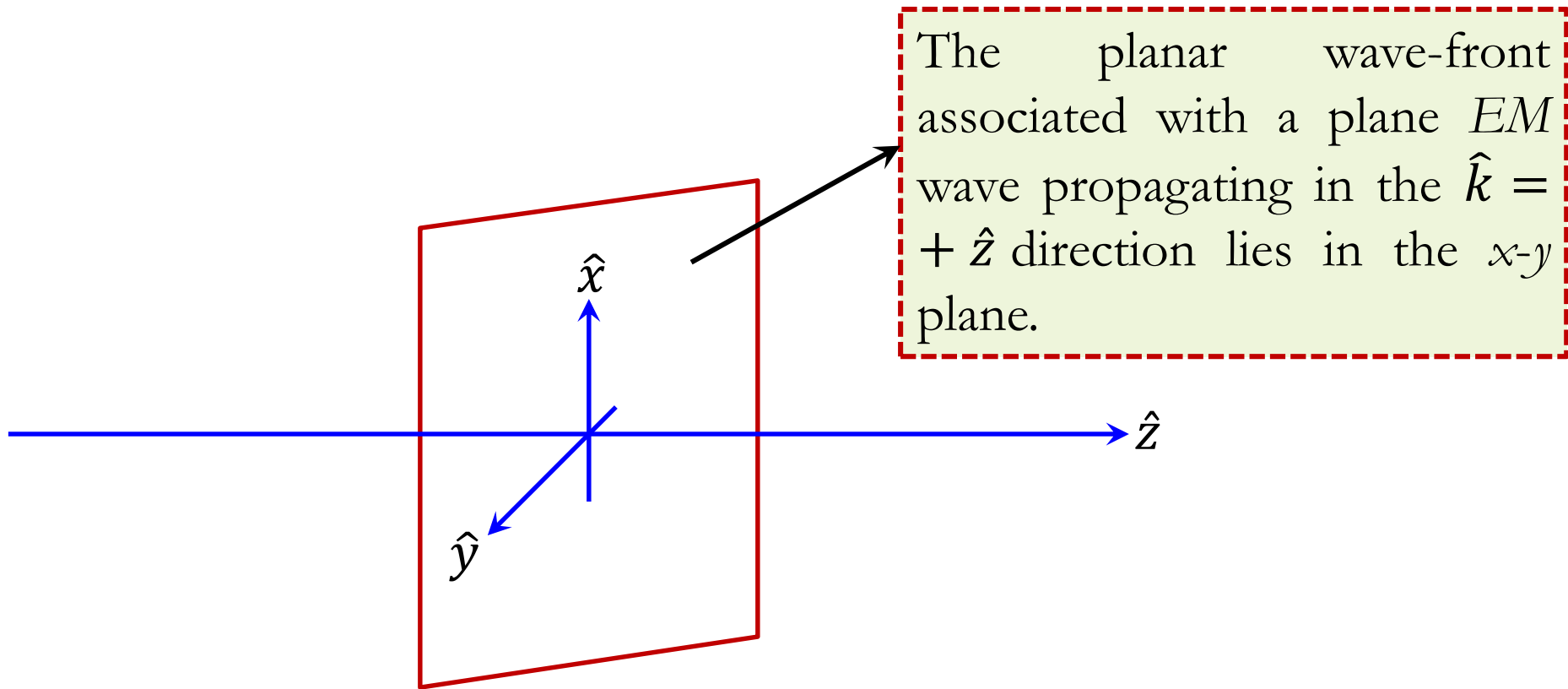
Monochromatic Plane Waves

Monochromatic EM plane waves propagating in free space/vacuum are sinusoidal EM plane waves consisting of a **single frequency ‘ f or ω ’, wavelength ‘ λ ’, angular frequency ‘ ω ’, and wavenumber ‘ k ’. They propagate with speed of light ‘ c ’.**

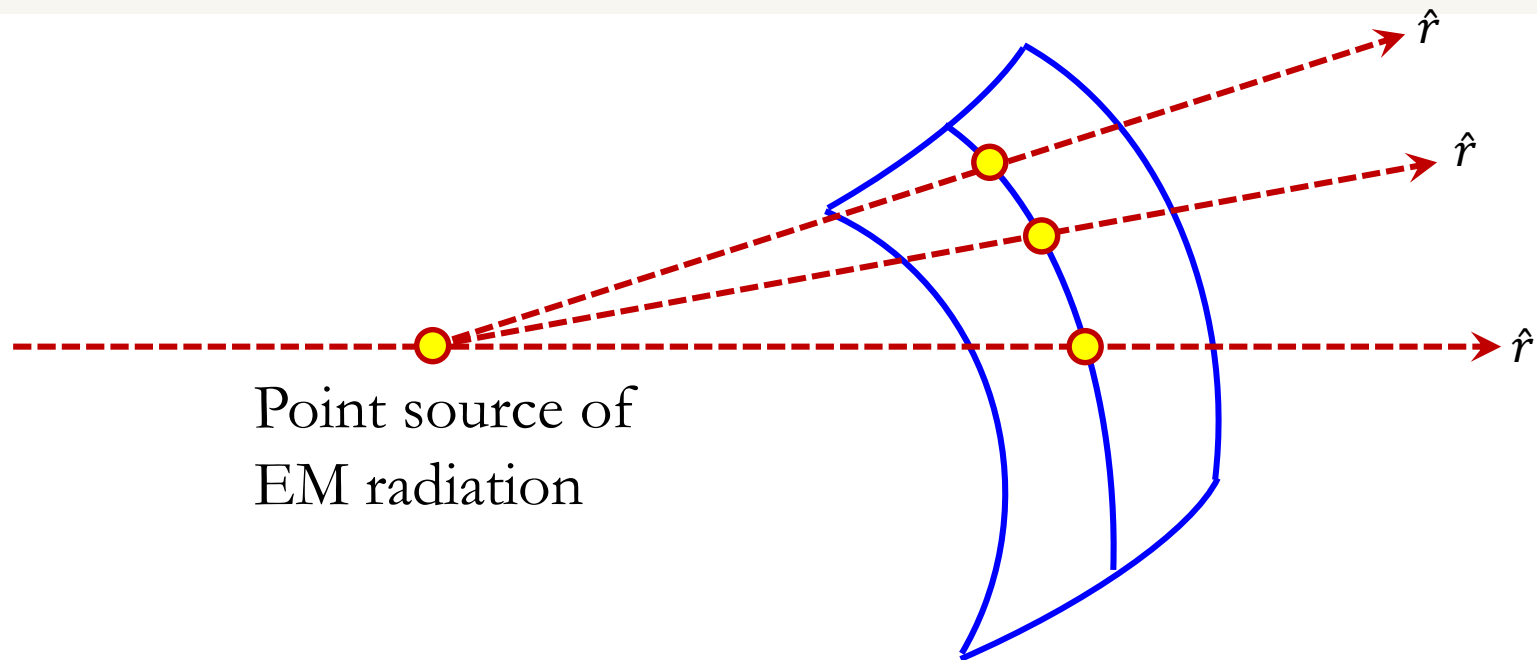
In the visible region of the EM spectrum $\{\sim 380 \text{ nm (violet)} \leq \lambda \leq \sim 780 \text{ nm (red)}\}$, EM light waves (consisting of real photons) of a given frequency / wavelength are perceived by the human eye as having a specific, single color. Hence we call such single-frequency, sinusoidal EM waves as **mono-chromatic**.

$$\tilde{B}_0 = \frac{1}{c} (\hat{k} \times \tilde{E}_0)$$

EM waves that propagate e.g. in the $+\hat{z}$ direction but which additionally have no explicit x - or y -dependence are known as plane waves, because for a given time, t the wave front(s) of the EM wave lie in a plane which is $\perp$ to the $\hat{z}$ -axis, as shown in the figure below:



Note that there also exist spherical *EM* waves – e.g. emitted from a point source, such as an atom, a small antenna or a pinhole aperture – the wavefronts associated with these *EM* waves are spherical, and thus do **not** lie in a **plane** to the direction of propagation of the *EM* wave:



Note: If the point source is **infinitely** far away from observer, then a spherical wave $\rightarrow$ plane wave. In this limit, the radius of curvature $R_c \rightarrow \infty$. i.e. a spherical surface becomes planar as $R_c \rightarrow \infty$.

A criterion for a {good} approximation of spherical wave as a plane wave is: $\lambda \ll R_c$

We are interested in fields of the form (**Monochromatic Waves**)

$$\tilde{E}(z, t) = \tilde{E}_0 e^{i(kz - \omega t)}$$

Propagating in the $\hat{k} = +\hat{z}$ direction

$$\tilde{B}(z, t) = \tilde{B}_0 e^{i(kz - \omega t)}$$

Propagating in the $\hat{k} = +\hat{z}$ direction

Here $\tilde{E}_0$ and $\tilde{B}_0$ are the (complex) amplitudes (the physical fields, of course, are the real parts of $\mathbf{E}$; and $\mathbf{B}$).

Now, the wave equations for $\mathbf{E}$ and $\mathbf{B}$ were derived from Maxwell's equations.

In particular, since $\nabla \cdot \vec{E} = 0$ and $\nabla \cdot \vec{B} = 0$, it follows that

$$(\tilde{E}_0)_z = (\tilde{B}_0)_z$$

Explanation given in next slide

$$\nabla = \frac{\partial}{\partial x} \hat{x} + \frac{\partial}{\partial y} \hat{y} + \frac{\partial}{\partial z} \hat{z}$$

In Cartesian co-ordinates

Thus $\nabla \cdot \vec{E} = 0$

$$\left(\frac{\partial}{\partial x} \hat{x} + \frac{\partial}{\partial y} \hat{y} + \frac{\partial}{\partial z} \hat{z} \right) \cdot (\tilde{E}_0 e^{i(kz - \omega t)}) = 0$$

$$\nabla \cdot \vec{E} = 0$$



1

$$\tilde{E}_0 = (E_{0x} \hat{x} + E_{0y} \hat{y} + E_{0z} \hat{z}) e^{i\delta}$$

E_{0x} , E_{0y} , E_{0z} = Amplitudes (constants) of the electric field components in x, y, z directions respectively.

$$\left(\frac{\partial}{\partial x} \hat{x} + \frac{\partial}{\partial y} \hat{y} + \frac{\partial}{\partial z} \hat{z} \right) \cdot (E_{0x} \hat{x} + E_{0y} \hat{y} + E_{0z} \hat{z}) e^{i\delta} e^{i(kz - \omega t)} = 0$$

Lets check each component separately

$$\frac{\partial}{\partial x} \hat{x} \cdot E_{0x} \hat{x} e^{i\delta} e^{i(kz-\omega t)} = 0$$

has no explicit x -dependence

These are just Magnitudes

$$\tilde{E}(z,t) = \tilde{E}_0 e^{i(kz-\omega t)}$$

$$\frac{\partial}{\partial y} \hat{y} \cdot E_{0y} \hat{y} e^{i\delta} e^{i(kz-\omega t)} = 0$$

has no explicit y -dependence

However $\frac{\partial}{\partial z} e^{az} = a e^{az}$

$$\tilde{E}_0 = (E_{0x} \hat{x} + E_{0y} \hat{y} + E_{0z} \hat{z}) e^{i\delta}$$

Thus

$$\frac{\partial}{\partial z} \hat{z} \cdot E_{0z} \hat{z} e^{i\delta} e^{i(kz-\omega t)} = ik E_{0z} e^{i(kz-\omega t)} e^{i\delta} = 0$$

true iff

$$E_{0z} = 0$$

Similarly for magnetic field **B**

$$\frac{\partial}{\partial z} \hat{z} \cdot B_{0z} \hat{z} e^{i\delta} e^{i(kz-\omega t)} = ik B_{0z} e^{i(kz-\omega t)} e^{i\delta} = 0$$

true iff

$$B_{0z} = 0$$

Thus, Maxwell's equations additionally tell us/impose the restriction that an electromagnetic plane wave cannot have any z- component of E or B to (or anti-parallel to) the propagation direction (in this case here, the $\hat{\mathbf{k}} = +\hat{\mathbf{z}}$ direction)

$$\left(\tilde{E}_0\right)_z = \left(\tilde{B}_0\right)_z = 0$$

Another way of stating this is that an EM plane wave **cannot** have any **longitudinal** components of E and B (i.e. components of E and B lying along the propagation direction).

Thus, Maxwell's equations additionally tell us that an EM plane wave is a purely transverse wave (at least while it is propagating in free space) – i.e. the components of E and B must be to $\perp$ propagation direction.

Furthermore: Maxwell's equations impose **yet another** restriction on the allowed form of $\mathbf{E}$ and $\mathbf{B}$ for an EM wave:

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{B} = \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t}$$

$$\text{Re}(\vec{\nabla} \times \vec{E}) = -\text{Re}\left(\frac{\partial \vec{B}}{\partial t}\right)$$

$$\text{Re}(\vec{\nabla} \times \vec{B}) = \text{Re}\left(\frac{1}{c^2} \frac{\partial \vec{E}}{\partial t}\right)$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{B} = \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t}$$

Lets find Curl of E

$$\begin{aligned}\vec{\nabla} \times \vec{E} &= \left(\frac{\partial \tilde{E}_z}{\partial y} - \frac{\partial \tilde{E}_y}{\partial z}\right) \hat{x} + \left(\frac{\partial \tilde{E}_x}{\partial z} - \frac{\partial \tilde{E}_z}{\partial x}\right) \hat{y} + \left(\frac{\partial \tilde{E}_y}{\partial x} - \frac{\partial \tilde{E}_x}{\partial y}\right) \hat{z} \\ &= -\frac{\partial \tilde{B}_x}{\partial t} \hat{x} - \frac{\partial \tilde{B}_y}{\partial t} \hat{y} - \frac{\partial \tilde{B}_z}{\partial t} \hat{z}\end{aligned}$$

These terms will become zero; since E is only z-dependent

$$\begin{aligned}\vec{\nabla} \times \vec{\tilde{E}} &= \left(\frac{\partial \tilde{E}_z}{\partial y} - \frac{\partial \tilde{E}_y}{\partial z} \right) \hat{x} + \left(\frac{\partial \tilde{E}_x}{\partial z} - \frac{\partial \tilde{E}_y}{\partial x} \right) \hat{y} + \left(\frac{\partial \tilde{E}_y}{\partial x} - \frac{\partial \tilde{E}_x}{\partial y} \right) \hat{z} \\ &= -\frac{\partial \tilde{B}_x}{\partial t} \hat{x} - \frac{\partial \tilde{B}_y}{\partial t} \hat{y} - \frac{\partial \tilde{B}_z}{\partial t} \hat{z}\end{aligned}$$

With:

$$\vec{\tilde{E}} = \tilde{E}_x \hat{x} + \tilde{E}_y \hat{y} + \tilde{E}_z \hat{z} = (E_{0x} \hat{x} + E_{0y} \hat{y} + \underset{\substack{\uparrow \\ \text{Zero}}}{\tilde{E}_z} \hat{z}) e^{i(kz - \omega t)} e^{i\delta}$$

$\underset{\substack{\uparrow \\ \text{Zero}}}{\tilde{E}_z}$

Thus:

$$\vec{\tilde{E}} = \tilde{E}_x \hat{x} + \tilde{E}_y \hat{y} = (E_{0x} \hat{x} + E_{0y} \hat{y}) e^{i(kz - \omega t)} e^{i\delta}$$

And similarly:

$$\vec{\tilde{B}} = \tilde{B}_x \hat{x} + \tilde{B}_y \hat{y} = (B_{0x} \hat{x} + B_{0y} \hat{y}) e^{i(kz - \omega t)} e^{i\delta}$$

Lets See how the below terms equals to zero

$$\left. \begin{array}{l} \frac{\partial \tilde{E}_z}{\partial y} \\ \frac{\partial \tilde{E}_y}{\partial x} \\ \frac{\partial \tilde{E}_x}{\partial y} \end{array} \right\} = 0$$

$$\vec{E} = (E_0)e^{i(kz-\omega t)}$$

$$\vec{E} = (E_{0x}\hat{x} + E_{0y}\hat{y} + E_{0z}\hat{z})e^{i(kz-\omega t)}$$

Writing the Individual components

So as the phasors

$$\vec{E}_x = (E_{0x})e^{i(kz-\omega t)} \rightarrow \frac{\partial E_x}{\partial y} = 0 \quad \frac{\partial \tilde{E}_x}{\partial y} = 0$$

$$\vec{E}_y = (E_{0y})e^{i(kz-\omega t)} \rightarrow \frac{\partial E_y}{\partial x} = 0 \quad \frac{\partial \tilde{E}_y}{\partial x} = 0$$

$$\vec{E}_z = (E_{0z})e^{i(kz-\omega t)} \rightarrow \frac{\partial E_z}{\partial y} = 0 \quad \frac{\partial \tilde{E}_z}{\partial y} = 0$$

$$\vec{E} = \tilde{E}_x\hat{x} + \tilde{E}_y\hat{y} + \tilde{E}_z\hat{z} = (E_{0x}\hat{x} + E_{0y}\hat{y} + E_{0z}\hat{z})e^{i(kz-\omega t)} e^{i\delta}$$

$$\frac{\partial \tilde{E}_y}{\partial z} = ikE_y$$

$$\vec{E}_y = (E_{0y})e^{i(kz-\omega t)}$$

$$\frac{\partial \vec{E}_y}{\partial z} = \frac{\partial}{\partial z} [(E_{0y})e^{i(kz-\omega t)}]$$



$$\frac{\partial \vec{E}_y}{\partial z} = (E_{0y}) \frac{\partial}{\partial z} [e^{i(kz-\omega t)}]$$

$$\frac{\partial \vec{E}_y}{\partial z} = (E_{0y})ik[e^{i(kz-\omega t)}]$$



$$\frac{\partial \vec{E}_y}{\partial z} = ik(E_{0y})[e^{i(kz-\omega t)}]$$

$$\frac{\partial \vec{E}_y}{\partial z} = ik\vec{E}_y$$



$$\frac{\partial \tilde{E}_y}{\partial z} = ik\tilde{E}_y$$

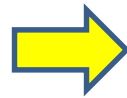
Time derivative of B :

$$\vec{B}_x = (B_{0x})e^{i(kz-\omega t)}$$

$$\frac{\partial \vec{B}_x}{\partial t} = -i\omega(B_{0x})e^{i(kz-\omega t)}$$

$$\frac{\partial \vec{B}_x}{\partial t} = -i\omega \vec{B}_x$$

$$\frac{\partial \tilde{E}_y}{\partial z} \hat{x} = \frac{\partial \tilde{B}_x}{\partial t} \hat{x}$$



$$ik\tilde{E}_y = -i\omega\tilde{B}_x$$

$$ik(E_{0y})e^{i(kz-\omega t)} e^{i\delta} = -i\omega(B_{0x})e^{i(kz-\omega t)} e^{i\delta}$$

$$ikE_{0y} = -i\omega B_{0x}$$

$$ikE_{0y} = -i\omega B_{0x}$$

1

Similarly

$$ikE_{0x} = +i\omega B_{0y}$$

2

$$\vec{\nabla} \times \vec{\tilde{E}} = \left(-\frac{\partial \tilde{E}_y}{\partial z} \right) \hat{x} + \left(\frac{\partial \tilde{E}_x}{\partial z} \right) \hat{y} = -\frac{\partial \tilde{B}_x}{\partial t} \hat{x} - \frac{\partial \tilde{B}_y}{\partial t} \hat{y}$$

$$\vec{\nabla} \times \vec{\tilde{B}} = \left(-\frac{\partial \tilde{B}_y}{\partial z} \right) \hat{x} + \left(\frac{\partial \tilde{B}_x}{\partial z} \right) \hat{y} = \frac{1}{c^2} \frac{\partial \tilde{E}_x}{\partial t} \hat{x} + \frac{1}{c^2} \frac{\partial \tilde{E}_y}{\partial t} \hat{y}$$

This can only be satisfied/can only be true **iff** the $\hat{x}$ and $\hat{y}$ relations are separately/independently satisfied $\forall (r, t)$

This implies

$$\frac{\partial \tilde{E}_y}{\partial z} \hat{x} = \frac{\partial \tilde{B}_x}{\partial t} \hat{x}$$

$$\frac{\partial \tilde{E}_x}{\partial z} \hat{y} = -\frac{\partial \tilde{B}_y}{\partial t} \hat{y}$$

$$ikE_{0y} = -i\omega B_{0x}$$

$$ikE_{0x} = +i\omega B_{0y}$$

1

2

The analysis for Curl of $\vec{\tilde{E}}$ leads the below set of equations

$$\vec{\nabla} \times \vec{\tilde{E}} = -\frac{\partial \vec{\tilde{B}}}{\partial t}$$

$$\frac{\partial \tilde{E}_y}{\partial z} \hat{x} = \frac{\partial \tilde{B}_x}{\partial t} \hat{x}$$

$$ikE_{0y} = -i\omega B_{0x}$$

1

$$\frac{\partial \tilde{E}_x}{\partial z} \hat{y} = -\frac{\partial \tilde{B}_y}{\partial t} \hat{y}$$

$$ikE_{0x} = +i\omega B_{0y}$$

2

Similarly for Curl of B

$$\vec{\nabla} \times \vec{\tilde{B}} = \frac{1}{c^2} \frac{\partial \vec{\tilde{E}}}{\partial t}$$

$$-\frac{\partial \tilde{B}_y}{\partial z} \hat{x} = \frac{1}{c^2} \frac{\partial \tilde{E}_x}{\partial t} \hat{x}$$

$$-ikB_{0y} = -\frac{1}{c^2} i\omega E_{0x}$$

3

$$\frac{\partial \tilde{B}_x}{\partial z} \hat{y} = \frac{1}{c^2} \frac{\partial \tilde{E}_y}{\partial t} \hat{y}$$

$$ikB_{0x} = -\frac{1}{c^2} i\omega E_{0y}$$

4

Comparison

$$ikE_{0y} = -i\omega B_{0x}$$

1

$$ikE_{0x} = +i\omega B_{0y}$$

2

$$-ikB_{0y} = -\frac{1}{c^2} i\omega E_{0x}$$

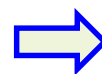
3

$$ikB_{0x} = -\frac{1}{c^2} i\omega E_{0y}$$

4

From 1

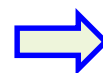
$$ikE_{0y} = -i\omega B_{0x}$$



$$E_{0y} = -\frac{\omega}{k} B_{0x}$$

From 2

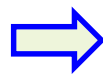
$$ikE_{0x} = +i\omega B_{0y}$$



$$E_{0x} = \frac{\omega}{k} B_{0y}$$

From 3

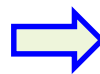
$$-ikB_{0y} = -\frac{1}{c^2} i\omega E_{0x}$$



$$B_{0y} = \frac{1}{c^2} \left(\frac{\omega}{k} \right) E_{0x}$$

From 4

$$ikB_{0x} = -\frac{1}{c^2} i\omega E_{0y}$$



$$B_{0x} = -\frac{1}{c^2} \left(\frac{\omega}{k} \right) E_{0y}$$

$$\text{Wave number (k)} = \frac{2\pi}{\lambda}$$

$$\frac{\omega}{k} = \frac{2\pi f \lambda}{2\pi} = f \lambda = c$$



Monochromatic waves travels at a speed of 'c'

From $\vec{\nabla} \times \vec{E}$ we got from the previous slide

$$E_{0y} = -\frac{\omega}{k} B_{0x}$$



$$B_{0x} = -\frac{1}{c} E_{0y}$$

$$E_{0x} = \frac{\omega}{k} B_{0y}$$



$$B_{0y} = \frac{1}{c} E_{0x}$$

From $\vec{\nabla} \times \vec{B}$ we got

$$\frac{\omega}{k} = \frac{2\pi f \lambda}{2\pi} = f \lambda = c$$

$$B_{0y} = \frac{1}{c^2} \left(\frac{\omega}{k} \right) E_{0x}$$



$$B_{0y} = \frac{1}{c} E_{0x}$$

$$B_{0x} = -\frac{1}{c^2} \left(\frac{\omega}{k} \right) E_{0y}$$



$$B_{0x} = -\frac{1}{c} E_{0y}$$

The same results which we got from $\vec{\nabla} \times \vec{E}$

So we really/actually only have two independent relations:

$$B_{0x} = -\frac{1}{c} E_{0y}$$

$$B_{0y} = \frac{1}{c} E_{0x}$$

Direction wise comparison $\rightarrow \hat{z} \times \hat{y} = -\hat{x}$

$\hat{z} \times \hat{x} = \hat{y}$

$$B_{0x} = -\frac{1}{c} E_{0y}$$

$$B_{0y} = \frac{1}{c} E_{0x}$$

Direction wise comparison $\rightarrow \hat{z} \times \hat{y} = -\hat{x}$

$\hat{z} \times \hat{x} = \hat{y}$

We can write the above two relations succinctly/compactly with ***one*** relation:

$$\frac{\omega}{k} = \frac{2\pi f \lambda}{2\pi} = f \lambda = c$$

$$\tilde{B}_0 = \frac{k}{\omega} (\hat{z} \times \tilde{E}_0)$$

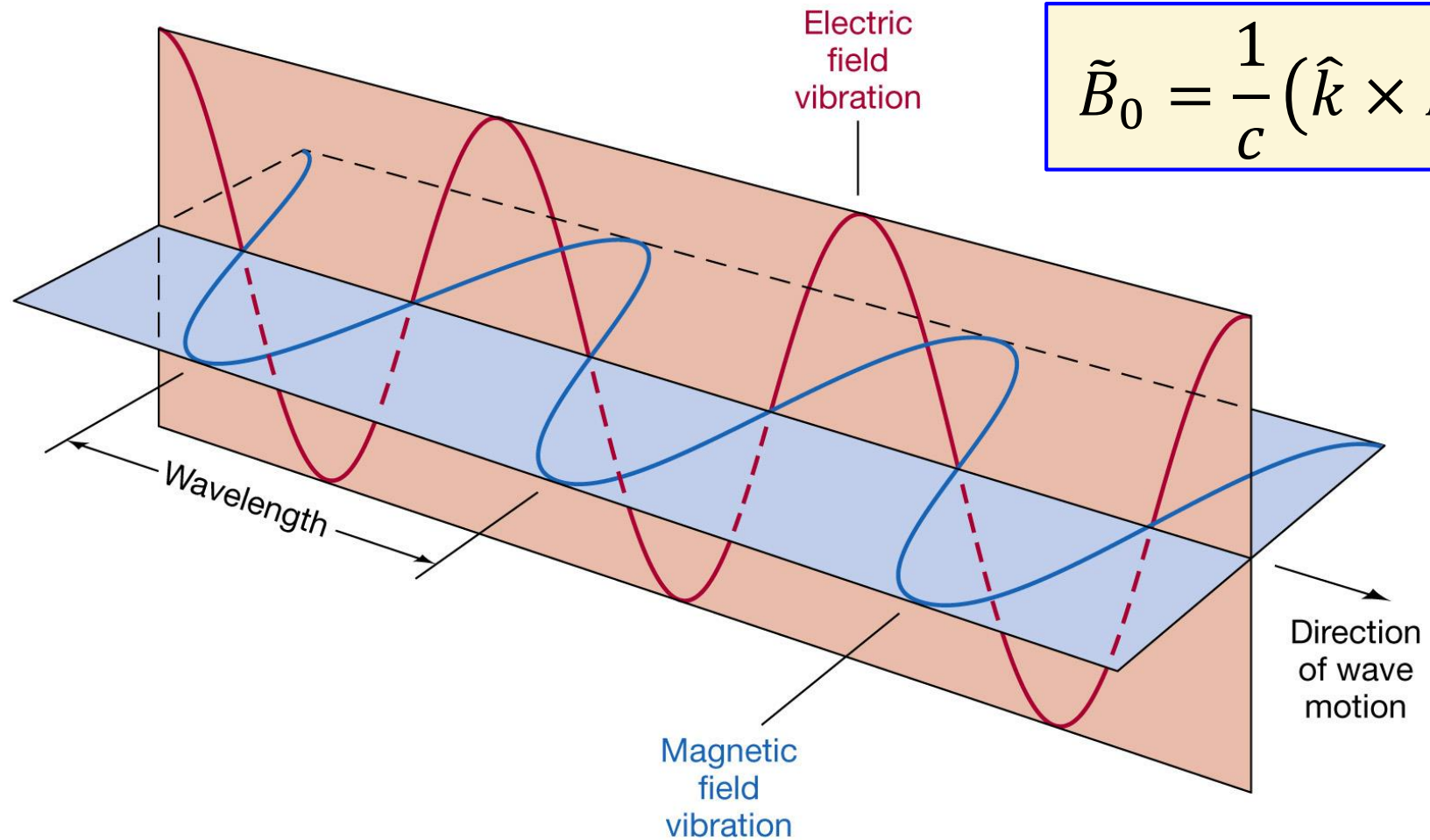
$$\tilde{B}_0 = \frac{1}{c} (\hat{k} \times \tilde{E}_0)$$

Physically, the mathematical relation $\tilde{\mathbf{B}}_0 = \frac{1}{c} (\hat{\mathbf{k}} \times \tilde{\mathbf{E}}_0)$ tells us that for a monochromatic *EM* plane wave propagating in free space, $\mathbf{E}$ and $\mathbf{B}$ are:

- a) In phase with each other.
- b) mutually perpendicular to each other and each is perpendicular to the propagation direction: $(\vec{\mathbf{E}} \perp \vec{\mathbf{B}}) \perp \hat{\mathbf{k}}$;
 $\hat{\mathbf{k}} = +\hat{\mathbf{z}}$ (propagation direction)

The $\mathbf{E}$ and $\mathbf{B}$ fields associated with this monochromatic plane EM wave are purely transverse.

Light = Electromagnetic "waves"



$$\tilde{B}_0 = \frac{1}{c} (\hat{k} \times \tilde{E}_0)$$

Transverse EM-waves: E & B variations are perpendicular to the direction of propagation

The purely real physical amplitudes of $\mathbf{E}$ and $\mathbf{B}$ are {also} related to each other by:

$$B_0 = \frac{1}{c} (E_0)$$

$$\tilde{B}_0 = \sqrt{B_{0x}^2 + B_{0y}^2}$$

$$\tilde{E}_0 = \sqrt{E_{0x}^2 + E_{0y}^2}$$

Generalization for Propagation of Monochromatic Plane EM Waves in an Arbitrary Direction

Obviously, there is nothing special/profound with regard to plane *EM* waves propagating in a specific direction in free space/the vacuum. They can propagate in any direction. We can easily generalize the mathematical description for monochromatic plane *EM* waves traveling in an arbitrary direction as follows:

$$\vec{\tilde{E}}(\mathbf{r}, t) = \tilde{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \hat{\mathbf{n}}$$

$$\vec{\tilde{B}}(\mathbf{r}, t) = \frac{1}{c} \hat{\mathbf{k}} \times \vec{E}(\mathbf{r}, t) = \frac{1}{c} \tilde{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} (\vec{k} \times \hat{\mathbf{n}})$$

We must have $\vec{\tilde{B}}(r, t) \perp \vec{\tilde{E}}(r, t) \perp \hat{k}$

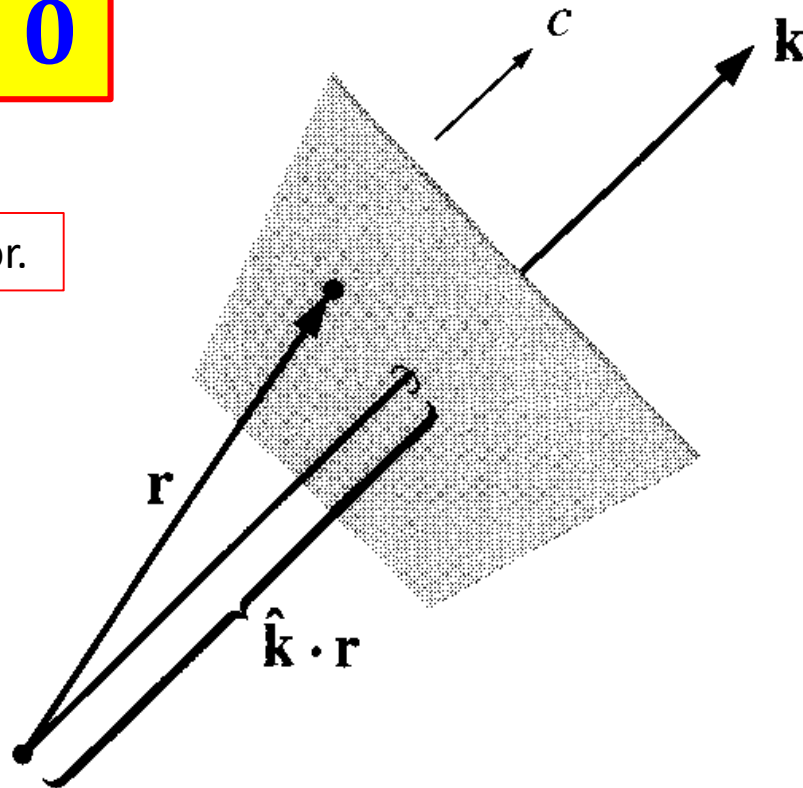
$$\vec{\tilde{E}} \cdot \vec{\tilde{B}} = 0$$

$$\vec{\tilde{E}} \cdot \hat{k} = 0$$

$$\vec{\tilde{B}} \cdot \hat{k} = 0$$

$$\hat{n} \cdot \hat{k} = 0$$

is the polarization vector.



The Direction of Propagation of a Monochromatic Plane EM Wave : $\hat{\mathbf{k}}$

The Real/Physical (Instantaneous) *EM* Fields are:

$$\vec{E}(r, t) = \text{Re} \left(\vec{\tilde{E}}(r, t) \right) = E_0 \cos(\vec{\mathbf{k}} \cdot \vec{\mathbf{r}} - \omega t + \delta) \hat{\mathbf{n}}$$

$\hat{\mathbf{n}} = \text{Polarization Vector } (\parallel \vec{E})$

$$\vec{B}(r, t) = \text{Re} \left(\vec{\tilde{B}}(r, t) \right) = B_0 \cos(\vec{\mathbf{k}} \cdot \vec{\mathbf{r}} - \omega t + \delta) (\hat{\mathbf{k}} \times \hat{\mathbf{n}})$$

$$B_0 = \frac{E_0}{c} \quad \text{In free space}$$

$$\vec{B}(r, t) = \text{Re} \left(\vec{\tilde{B}}(r, t) \right) = \frac{E_0}{c} \cos(\vec{\mathbf{k}} \cdot \vec{\mathbf{r}} - \omega t + \delta) (\hat{\mathbf{k}} \times \hat{\mathbf{n}})$$